

SARDAR PATEL UNIVERSITY

T.Y.B.C.A. (Fifth Semester) (CBCS) EXAMINATION – 2017

US05FBCA01 : Operations Research

Date: 07/11/2017, Tuesday

Time: 02:00 to 05:00 PM

Total Marks : 70

Q.1 Multiple choice questions:

[10]

1 In graphical representation the bounded region is known as _____ region.

[a] Solution

[b] feasible solution

[c] basic solution

[d] optimal

2 In LPP the condition to be satisfied is

[a] Constraints have to be linear

[b] Objective function have to be linear

[c] both [a] and [b].

[d] none of the above

3 The linear function of variables which is to be maximized or minimized is called _____.

[a] constraints

[b] basic requirements

[c] objective function

[d] none of them

4 In the simplex method for solving of LPP number of variables can be _____.

[a] Not more than three

[b] at least three

[c] at least two

[d] none of them

5 The _____ variable is added to the constraint of less than equal to type.

[a] slack

[b] Surplus

[c] artificial

[d] basic

6 The Penalty in VAM represents difference between _____ cost of respective row/column.

[a] Two Largest

[b] smallest two

[c] largest and smallest

[d] none of them

7 In Assignment Problem the value of decision variable x_{ij} is _____.

[a] no restriction

[b] two or one

[c] one or zero

[d] none of them

8 If number of sources is not equal to number of destination in Assignment problem then it is called _____.

[a] unbalanced

[b] symmetric

[c] unsymmetric

[d] balanced

9 Burst and Merge are types of _____ in networking.

[a] event

[b] activity

[c] arrow

[d] tools

10 Activity which is completed before starting new activity is called _____.

[a] dummy

[b] Predecessor

[c] successor

[d] none of them

Q.2 Attempt any ten out of twelve.

[20]

1. Define i] Unbounded solution ii] Optimum solution
2. Define Operation research.
3. Give any four models of operations research.
4. When is Big M method useful?
5. What is the condition for optimality in simplex table?
6. Differentiate graphical method and simplex method in LPP.
7. What is the Assignment problem?
8. What is travelling salesman problem?
9. What is transportation problem?
10. In brief explain problem of sequencing.
11. Define two types of events used in network analysis.
12. What is successor activity?

Q.3 [a] Note down the applications of operations research.

[4]

Q.3[b] A firm manufactures two types of products A and B and sells them at a profit of Rs. 200 on type A and Rs. 300 on type B. each product is processed on two machines G and H. type A requires 1 minute of processing time on G and 2 minutes on H; Type B requires 1 minute on G and 1 minute on H. the machine G is available for not more than 6 hours, 40 minutes while H is available for 10 hours during any working day. Formulate this problem as a linear programming problem.

[6]

OR

Q.3[a] Note down the phase of operations research.

[4]

Q.3[b] Solve the below LPP using graphical method.

[6]

$$\text{Maximize } Z = 40X_1 + 30X_2$$

Subject to:

$$3X_1 + X_2 \leq 30, X_1 \leq 8, X_2 \leq 12$$

$$X_1, X_2 \geq 0$$

Q.4 [a] Write the standard form of LPP for the following LPP

[4]

$$\text{Maximize } Z = 13x_1 + 25x_2$$

$$\text{Subject to } 21x_1 + 3x_2 \leq 40, \quad 5x_1 + 2x_2 \leq 7, \quad x_1, x_2 \geq 0$$

Q.4[b] Solve the below LPP using Big-M method.

[6]

$$(1) \text{Maximize } Z = 3x_1 - x_2$$

$$\text{Subject to } 2x_1 + x_2 \geq 2, \quad x_1 + 3x_2 \leq 3, \quad x_1, x_2 \geq 0$$

OR

Q.4[a] Define slack and surplus variables with example.

[4]

Q.4[b] Solve the below LPP using simplex method.

[6]

$$\text{Max } Z = 2X_1 + 4X_2$$

Subject to:

$$X_1 + 2X_2 \leq 5, \quad X_1 + X_2 \leq 4,$$

$$X_1, X_2 \geq 0$$

Q.5 [a] What is difference between transportation problem and assignment problem? [4]

Q.5[b] Obtain the initial solution of below TP using Vogel's approximation method. [6]

	I	II	III	IV	Supply
A	21	16	15	13	11
B	17	18	14	23	13
C	32	27	18	41	19
Demand	6	10	12	15	

OR

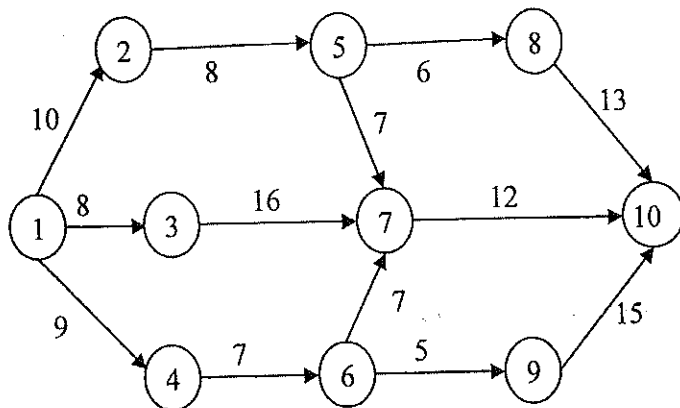
Q.5[a] Give the algorithm of LCM to obtain basic feasible initial solution to transportation problem. [4]

Q.5[b] Solve the below assignment problem. [6]

	P	Q	R	S
A	22	30	21	15
B	18	33	9	31
C	44	25	24	21
D	23	30	28	14

Q.6 [a] Write down the procedure for solving problem of sequencing with two machines. [4]

[b] Find the critical path and calculate the Total float and Free float for the following PERT diagram. [6]



OR

Q.6[a] State the rules for drawing network diagram. [4]

Q.6[b] In a machine shop 8 different products are being manufactured each requiring time on two different machines A and B are given in the table below: [6]

Product	1	2	3	4	5	6	7	8
Machine-A	30	45	15	20	80	120	65	10
Machine B	20	30	50	35	35	40	50	20

Find an optimal sequence of processing of different product in order to minimize the total manufactured time for all product. Find total ideal and elapsed time.

101

...the ... of ...

102

...the ... of ...

103

...the ... of ...

104

...the ... of ...

105

...the ... of ...

106

...the ... of ...

...the ... of ...